

Moduli of Langlands parameters

David Helm, Xiangqian Yang

Lecture 1 (DH) $F \mid \mathcal{O}_p$, G quasi-split conn. red. gp/ F

$\hat{G}$ dual gp, ${}^L G = \hat{G} \rtimes W_F$ (makes sense over $\mathbb{Z}$)

Def. A Langlands parameter for G over $\overline{\mathcal{O}_e}$ is an l -adically cts $\rho: W_F \rightarrow {}^L G(\overline{\mathcal{O}_e})$

s.t. $W_F \rightarrow {}^L G(\overline{\mathcal{O}_e})$
 $\Downarrow$
 W_F (up to conj. by $\hat{G}$)

$R = \mathbb{Z}[\frac{1}{p}]$ -algebra

Ideas: "discretize" W_F

$$W_F / P_F \simeq \mathbb{Z} \rtimes I_F \simeq \mathbb{Z} \rtimes \prod_{l \neq p} \mathbb{Z}_l$$

$\downarrow \quad \quad \downarrow$
 $F_r \quad \quad S$

$$F_r s F_r^{-1} = s^q$$

Set $\overline{W}_F' \subset W_F / P_F$ subgp gen'd by F_r, s . $\mathbb{Z} \rtimes \mathbb{Z}[\frac{1}{p}]$

W_F' = preimage of $\overline{W}_F'$ in W_F .

basis of open subgroups of P_F :

Define: $\tilde{\text{Par}}_G^{(R)} := \{ \rho': W_F' \rightarrow {}^L G(R) : \rho'|_{P_F} \text{ has open kernel} \}$

$\searrow \quad \downarrow$
 $\text{Par}_G = [\tilde{\text{Par}}_G / \hat{G}]$ $\tilde{\text{Par}}_G$ depends on F_r, s (but $\tilde{\text{Par}}_G \otimes \mathbb{Z}_l$ does not!)

Th (Scholze) Par_G indep of choices.

—

Assume G splits over a tame ext'n of F .

$\tilde{\text{Par}}_G^{\text{tame}}$ = closed subscheme of $\tilde{\text{Par}}_G$

$$\left\{ P: P|_{P_F} = \text{id} \right\} \quad \hookrightarrow \hat{G}\text{-stable}$$

$$P(w) = \text{id} \times w, \quad w \in P_F$$

$$= \left\{ (F, \sigma) : F \in \hat{G} \times F_r, \quad \sigma \in \hat{G} \times S, \quad F \sigma F^{-1} = \sigma^q \right\}$$

$$\psi: \hat{G} \times \hat{G} \rightarrow \hat{G} \quad (g \times F_r)(g' \times S)(g \times F_r)^{-1}$$

$$= \psi(g, g') (g' \times S)^q$$

$$\tilde{\text{Par}}_G^{\text{tame}} = \psi^{-1}(\text{id})$$

Prop. $\dim \tilde{\text{Par}}_G^{\text{tame}} = \dim \hat{G} + 1$. In particular, $\tilde{\text{Par}}_G^{\text{tame}}$ is an lci.

Idea: $\hat{G}$ -conj. class of σ is constrained.

(Only finitely many)

Prop $\tilde{\text{Par}}_G^{\text{tame}}$ is generically reduced

$$\text{Compute: } \dim T_P \tilde{\text{Par}}_G^{\text{tame}} = \dim \hat{G} + \dim H^0(W_F^1, \text{ad}_P(1))$$

Upshot: $\tilde{\text{Par}}_G^{\text{tame}}$ is reduced + flat over $\mathbb{Z}[\frac{1}{p}]$!

$$\rho^{\text{univ}} : W_F^1 / P_F \rightarrow L_G(\mathcal{O}_{\tilde{\text{Par}}_G^{\text{tame}}}) \text{ universal rep.}$$

$$\rho^{\text{univ}}(s) : \exists N \text{ (dep. on } G) : \forall k, \forall P \in \tilde{\text{Par}}_G^{\text{tame}}(k), \quad P(s)^N \text{ is unipotent.}$$

$\Rightarrow \rho^{\text{univ}}(s)^N \in \mathcal{O}_{\widetilde{\text{Par}}_G^{\text{tame}}} - \text{pts of unipotent cone of } \widehat{G} \times S^N$

$\Rightarrow$ over $\mathbb{Z}_\ell$, ρ^{univ} extends to $\rho^{\text{univ}}: W_F \rightarrow {}^L G(\mathcal{O}_{\widetilde{\text{Par}}_G^{\text{tame}}})$

Def. A $\mathbb{Z}_\ell$ -alg R is ℓ -adically separated if $\bigcap \ell^n R = \{0\}$.

For such R , $\rho: W_F \rightarrow {}^L G(R)$ is ℓ -adically cts if $\forall n$,

$\ker \rho_n: W_F \rightarrow {}^L G(R/\ell^n R)$ is open

Prop $\mathcal{O}_{\widetilde{\text{Par}}_{G, \mathbb{Z}_\ell}^{\text{tame}}}$ is ℓ -adically separated, and $\rho^{\text{univ}}: W_F \rightarrow {}^L G(R^{\text{tame}})$
 $\parallel$
 R^{tame} is ℓ -adically cts.

(Can extend to all R by saying $\rho: W_F \rightarrow {}^L G(R)$ is ℓ -adically cts if

ρ arises as $\rho' \otimes_{R'} R$, $\rho': W_F \rightarrow {}^L G(R')$ cts, R' ℓ -adically sep.

Th. (Cotner) $\widetilde{\text{Par}}_G^{\text{tame}}$ is conn'd. over $\overline{\mathbb{Z}}[\frac{1}{p}]$.

Th (DHKM) Let $\rho: W_F' \rightarrow {}^L G(\overline{\mathbb{A}}_1) \in \widetilde{\text{Par}}_G^P(\overline{\mathbb{A}}_1)$. $\widetilde{\text{Par}}_G^P$ conn cpt of

$\widetilde{\text{Par}}_G$ containing ρ , $\exists G_P$ tame conn'd red gp, finite gp scheme H , st.

$$\widetilde{\text{Par}}_G^P \simeq \widetilde{\text{Par}}_{G_P}^{\text{tame}} / H.$$

$\Rightarrow \widetilde{\text{Par}}_G^P$ is lci, $\dim = 1$, flat, reduced

Moreover, $\rho_P^{\text{univ}}: W_F' \rightarrow {}^L G(\mathcal{O}_{\widetilde{\text{Par}}_G^P})$ extends uniquely over $\mathbb{Z}_\ell$ to W_F .

When G is tame and $p \nmid |W(G)|$, then $H = \{\text{id}\}$.

Lecture 2 (XY) E is a local field, G quasi-split conn. red. gp / E

$\leadsto \hat{G}$ pinned dual group / $\mathbb{Z}_\ell$ $\hat{T} \subset \hat{B} \subset \hat{G} \hookrightarrow W_E \twoheadrightarrow Q$ finite quotient

Recall. $\tilde{\text{Par}}_G = Z^1(W_E, \hat{G}) / \mathbb{Z}_\ell$

$$R/\mathbb{Z}_\ell \mapsto \left\{ \varphi: W_E \rightarrow \hat{G}(R) \rtimes Q \text{ "continuous" } \right\}$$

$$\searrow \quad \begin{matrix} \downarrow \\ Q \end{matrix}$$

Last time. $Z^1(W_E, \hat{G})$ is a disj. union of affine schemes, locally of f.t. / $\mathbb{Z}_\ell$,

lci.

flat of dim $\dim \hat{G}$ over $\mathbb{Z}_\ell$

$$\text{Par}_G = Z^1(W_E, \hat{G}) / \hat{G}$$

§ Coarse moduli space $Z^1(W_E, \hat{G}) // \hat{G}$ GIT quotient

Def. ($L | \mathbb{Z}_\ell$ alg closed field) parabolic subgp of $\hat{G}_L \rtimes Q := \hat{G}(L)$ -conj. of $\hat{P}_L \rtimes Q$
 $\hat{P}$ std parabolic stable under Q -action

$$\text{Levi subgp of } \hat{G}_L \rtimes Q := \text{---} \hat{M}_L \rtimes Q$$

$$\hat{M} \text{ std Levi}$$

Def $L/\mathbb{Z}_\ell$ $\varphi: W_E \rightarrow \hat{G}(L) \rtimes Q$ is semisimple if whenever $\text{im}(\varphi) \subset \text{parabolic subgp}$, it lies in a Levi subgp of it.

Th (DHKM) L , $\varphi: W_F \rightarrow \hat{G}(L) \rtimes Q$. TFAE:

(1) φ is semisimple

(2) the $\hat{G}_L$ -orbit of φ in $Z^1(W_E, \hat{G})$ is closed.
 Pages

Cor. $\left\{ L\text{-pts of } \mathbb{Z}^1(W_E, \hat{G}) // \hat{G} \right\} \leftrightarrow \left\{ \text{semisimple } L\text{-par. } / L \right\} / \hat{G}(L)\text{-conj.}$

$\xrightarrow{\quad}$

Excursion operators

Goal. Give a presentation of $\mathcal{O}(\mathbb{Z}^1(W_E, \hat{G}))^{\hat{G}}$.

Recall $\mathbb{Z}^1(W_E, \hat{G}) = \varinjlim_{\substack{P \subset P_F \\ \text{open}}} \mathbb{Z}^1(W_E/P, \hat{G})$

$\uparrow$ open closed

Fix $W \subset W_E/P$ discrete dense. $\mathbb{Z}^1(W, \hat{G}) \simeq \mathbb{Z}^1(W_E/P, \hat{G})$.

F_n free group of n -generators:

Given $F_n \rightarrow W \hookrightarrow \mathbb{Z}^1(W, \hat{G}) \rightarrow \mathbb{Z}^1(F_n, \hat{G}) \simeq \hat{G}^n$

$\hookrightarrow \mathcal{O}(\mathbb{Z}^1(F_n, \hat{G})) \rightarrow \mathcal{O}(\mathbb{Z}^1(W, \hat{G}))$

$\hookrightarrow \varinjlim_{n, F_n \rightarrow W} \mathcal{O}(\mathbb{Z}^1(F_n, \hat{G})) \rightarrow \mathcal{O}(\mathbb{Z}^1(W, \hat{G})).$

Claim. This is a $\hat{G}$ -equiv. isom.

Pf. $R/\mathbb{Z}_\ell, \quad \text{Hom} \left(\varinjlim_{n, F_n \rightarrow W} \mathcal{O}(\mathbb{Z}^1(F_n, \hat{G})), R \right)$

$= \lim_{n, F_n \rightarrow W} \text{Hom} \left(\mathcal{O}(\mathbb{Z}^1(F_n, \hat{G})), R \right)$

$= \lim_{n, F_n \rightarrow W} \left\{ F_n \rightarrow \hat{G}(R) \rtimes \mathcal{Q} \right\}$

$= \left\{ \varinjlim_{n, F_n \rightarrow W} F_n \rightarrow \hat{G}(R) \rtimes \mathcal{Q} \right\}$

$\parallel$
 W

□

Take $\hat{G}$ -inv.

$$\operatorname{colim}_{n, F_n \rightarrow W} \mathcal{O}(\mathbb{Z}^1(F_n, \hat{G}))^{\hat{G}} \longrightarrow \mathcal{O}(\mathbb{Z}^1(W, \hat{G}))^{\hat{G}}$$

!!

Def: $\operatorname{Exc}(W, \hat{G})$ alg of excursion operators.

Over $\mathcal{O}_\ell$, $(-)^{\hat{G}}$ commutes with colimits

$$\Rightarrow \operatorname{Exc}(W, \hat{G})_{\mathcal{O}_\ell} \xrightarrow{\sim} \mathcal{O}(\mathbb{Z}^1(W, \hat{G}))^{\hat{G}}$$

Th (FS) Assume $\ell \nmid |\pi_1(\hat{G})_{\text{tor}}|$, then $\operatorname{Exc}(W, \hat{G}) \xrightarrow{\sim} \mathcal{O}(\mathbb{Z}^1(W, \hat{G}))^{\hat{G}}$

Rem. $\operatorname{Exc}(W, \hat{G}) \rightarrow \mathcal{O}(\mathbb{Z}^1(W, \hat{G}))^{\hat{G}}$ univ. homeomorphism

$$\Rightarrow L \mid \mathbb{Z}_\ell, \quad \left\{ L\text{-pts of } \operatorname{Exc}(W, \hat{G}) \right\} = \left\{ \text{s.s. } L\text{-parameters} \right\} / \hat{G}(L)$$

Th. (V. Lafforgue, FS) $\mathcal{C}$ $\mathbb{Z}_\ell$ -linear cat,

$\forall I$ finite set, $\operatorname{Rep}(\mathcal{C}^I)$ - linear ^{monoidal} ~~Q~~-functors

$$\operatorname{Rep}((\hat{G} \rtimes \mathcal{Q})^I) \longrightarrow \operatorname{End}(\mathcal{C})^{B_{W^I}}$$

$$v \longmapsto T_v: \mathcal{C} \rightarrow \mathcal{C}$$

$$x \mapsto T_v(x) \in W^I$$

s.t. for any $I \rightarrow J$,

$$\operatorname{Rep}((\hat{G} \rtimes \mathcal{Q})^I) \longrightarrow \operatorname{End}(\mathcal{C})^{B_{W^I}}$$

$$\downarrow \quad \sim \quad \downarrow$$

$$\operatorname{Rep}((\hat{G} \rtimes \mathcal{Q})^J) \longrightarrow \operatorname{End}(\mathcal{C})^{B_{W^J}}$$

$$= (\hat{G}^J \rightarrow \hat{G}^I, W^J \rightarrow W^I)$$

Parsh

Then there is a nat'l map $\text{Exc}(W, \hat{G}) \rightarrow \text{End}(\text{id}_e) \leftarrow \text{Bernstein centre}$

Sketch assume $\mathbb{Q} = \{1\}$.

Def excursion datum $(I, V, \alpha, \beta, (r_i)_{i \in I}) = D$

- I finite set
- $V \in \text{Rep}(\hat{G}^I)$
- $\alpha: 1 \rightarrow V|_{\Delta \hat{G}}, \quad \beta: V|_{\Delta \hat{G}} \rightarrow 1$
- $(r_i)_{i \in I} \in W^I$

$$\rightsquigarrow t(V, \alpha, \beta) \in \mathcal{O}(\hat{G} \backslash \hat{G}^I / \hat{G})$$

$$t(V, \alpha, \beta) \left((g_i)_{i \in I} \right) = \left(1 \xrightarrow{\alpha} V \xrightarrow{(g_i)_{i \in I}} V \xrightarrow{\beta} 1 \right)$$

$\rightsquigarrow$ excursion operator

$$S_D: \left(\text{id}_e \xrightarrow{\alpha} T_V \xrightarrow{(r_i)_{i \in I}} T_V \xrightarrow{\beta} \text{id}_e \right) \in \text{End}(\text{id}_e)$$

For $F_n \xrightarrow{(r_i)_{i \in I, n \in \mathbb{N}}} W$, define $\mathcal{O}(Z^1(F_n, \hat{G}))^{\hat{G}} = \mathcal{O}(\hat{G}^n)^{\hat{G}} = \mathcal{O}(\hat{G} \backslash \hat{G}^{\{0, \dots, n\}} / \hat{G})$

$$\begin{array}{ccc} t(V, \alpha, \beta) & & \\ \downarrow & & \downarrow \\ S_D & & \text{End}(\text{id}_e) \end{array}$$

$$\rightsquigarrow \text{Exc}(W, \hat{G}) \rightarrow \text{End}(\text{id}_e)$$

$$D = \{ (0, \dots, n), V, \alpha, \beta, (1, r_1, \dots, r_n) \}$$

Rem. $\underline{FS}: \text{Exc}(W, \hat{G}) \rightarrow \text{End}(\text{id}_{D(\text{Bun}_{\hat{G}})}) \rightarrow \text{End}(\text{id}_{\text{Rep}(G(E))})$

T_V are Hecke operators defined using geom. Satake.

$\Rightarrow \forall \pi$ irrep of $G(E)/L$, get $\text{Exc}(W, \hat{G}) \rightarrow \text{End}(\pi) = L$

$\rightsquigarrow$ semisimple L -parameter φ_π .

§. Weil-Deligne parameters $\text{coeff} = \overline{\mathbb{Q}_\ell}$.

Def. $R/\overline{\mathbb{Q}_\ell}$, A Weil-Deligne parameter is a pair (h, X)

$$- h: W_E \rightarrow \hat{G}(R) \rtimes \mathbb{Q}$$

$$\begin{array}{ccc} & \sim & \downarrow \\ & \searrow & \mathbb{Q} \end{array}$$

s.t. $h|_{I_E}$ factors through a finite quotient.

$$- X \in N_{\hat{G}}(R) \text{ s.t. } \text{Ad}_{h(r)}(X) = \|r\| \cdot X, \quad \| \cdot \|: W_E \rightarrow \overline{\mathbb{Q}_\ell}^\times \\ \text{for } r \in W_E$$

Such an $h: W_E \rightarrow \hat{G}(R) \rtimes \mathbb{Q}$ is a Weil parameter.

$\leadsto$ stack of Weil-Deligne par. Par_G^{WD}

Weil par. Par_G^W

$$\lambda: \text{Par}_G^{\text{WD}} \rightarrow \text{Par}_G^W$$

Th (Grothendieck's monodromy thm, Zhu) $\text{Par}_G^{\text{WD}} \simeq \text{Par}_{G, \overline{\mathbb{Q}_\ell}}$

Fix $Fr \in W_E$, $\tau \in I_E^t$ top generator.

$$- (h, X) \in \text{Par}_G^{\text{WD}}(\overline{\mathbb{Q}_\ell}) \longmapsto \varphi \in \text{Par}_G(\overline{\mathbb{Q}_\ell})$$

$$- \varphi(Fr) = h(Fr)$$

$$- \varphi(r) = h(r) \exp(|r|_\tau \cdot X), \quad r \in I_E$$

$$|\cdot|_\tau: I_E \rightarrow I_E^t \simeq \hat{\mathbb{Z}}^p(1) \xrightarrow{\tau} \mathbb{Z}_\ell$$

$$- \varphi \mapsto (h, X)$$

$$- h(Fr) = \varphi(Fr)$$

$$- h(r) = \varphi(r)^{ss}, \quad r \in I_E$$

$$- X = \frac{1}{m} \log \left(\varphi(\tau)^m \right)_{\text{p-adj}}$$

for $m \gg 0$, $\varphi(\tau)^m$ unipotent
 $\log: \mathcal{U}_{\hat{G}} \simeq \mathcal{N}_{\hat{G}}: \exp$

Def $h: W_E \rightarrow \hat{G}(\bar{A}_E) \rtimes Q$ Weil parameter

$$V_h := \lambda^{-1}(\{h\})_{\text{red}}$$

Vogan variety

$$\lambda: \text{Par}_{\hat{G}}^{\text{WD}} \rightarrow \text{Par}_{\hat{G}}^W$$

$$\begin{array}{ccc} \uparrow & \nearrow & \uparrow \\ V_h & & \{h\} \end{array}$$

$$V_h \simeq \hat{g}^{h(I_F)=1, h(Fr)=1} \subset \mathcal{V}_{\hat{G}}$$

$$\text{Then } V_h / C_{\hat{G}}(h) \hookrightarrow \text{Par}_{\hat{G}}^{\text{WD}} \simeq \text{Par}_{\hat{G}, \bar{A}_E}$$

Def (1) $\varphi: W_E \rightarrow \hat{G}(\bar{A}_E) \rtimes Q$ is essentially discrete if $C_{\hat{G}}(\varphi) / Z_{\hat{G}}^{W_E}$ is finite.

(2) φ is supercuspidal if φ ess. discrete and semisimple.

Fact. φ ess. discrete $\Rightarrow \varphi$ Frob ss.

Let $h = \varphi^{\text{ss}}$ Weil parameter.

Prop. $C = (Z_{\hat{G}}^{W_E})^0$, φ ess. discrete

$$C \times \{\varphi\} / C_{\hat{G}}(\varphi) \xrightarrow{\text{open}} C \times V_h / C_{\hat{G}}(h) \xrightarrow{\text{closed}} \text{Par}_{\hat{G}}$$

open

In particular, φ is a smooth pt of $\text{Par}_{\hat{G}}$.

Rem. φ s.c. $V_h = \{\varphi\} \Rightarrow C \times \{\varphi\} / C_{\hat{G}}(\varphi) \hookrightarrow \text{Par}_{\hat{G}}$ open closed.

Ex. $\hat{G} = \text{GL}_n$, $\varphi: W_E \rightarrow W_E^t \rightarrow \text{GL}_n(\bar{A}_E)$

$$\varphi(\text{Fr}) = \begin{pmatrix} q^{\frac{n-1}{2}} & & & \\ & q^{\frac{n-3}{2}} & & \\ & & \ddots & \\ & & & q^{\frac{1-n}{2}} \end{pmatrix}, X = \begin{pmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{pmatrix}$$

Steinberg parameter

$$V_h = \begin{pmatrix} 0 & * & 0 \\ & \ddots & * \\ 0 & & 0 \end{pmatrix} \simeq \mathbb{A}^{n-1}, C_{\hat{G}}(h) = \hat{T} = \mathbb{G}_m^n$$

$$\begin{array}{c}
 \rightsquigarrow G_m \times \{p\} / G_m \xrightarrow{\text{open}} G_m \times A^{n-1} / G_m \xrightarrow{\text{closed}} \text{Par}_{G_m} \\
 \underbrace{\hspace{10em}}_{\text{smooth}}
 \end{array}$$

GL₂ case Par_{GL_n}

$$\begin{array}{ccccc}
 \begin{array}{c} | \\ \circ \\ | \end{array} & \longrightarrow & | & \longrightarrow & \begin{array}{c} | \\ \hline | \end{array} \text{ wr } \\
 / G_m^2 & & / G_m^2 & & / G_m^2
 \end{array}$$